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% PI Controller Comparison: Direct Synthesis vs. Cohen-Coon
clc; clear; close all;

--- User-defined process parameters ---

```
K = 3;           % Process gain
tau = 2;         % Process time constant
theta = 1;       % Process dead time
```

--- Transfer function setup ---

```
s = tf('s');
Gp = K * exp(-theta * s) / (tau * s + 1); % FOPTD model
```

--- (a) Direct Synthesis Tuning ---

```
Kc_DS = tau / (K * (theta + tau));
PI_DS = Kc_DS * (1 + 1 / (tau * s));
```

--- (b) Cohen-Coon Tuning ---

```
Kc_CC = (tau / (K * theta)) * (0.9 + theta / (12 * tau));
tauI_CC = ((30 + 3 * (theta / tau)) / (9 + 20 * (theta / tau))) * theta;
PI_CC = Kc_CC * (1 + 1 / (tauI_CC * s));
```

--- Closed-loop systems ---

```
CL_DS = feedback(PI_DS * Gp, 1);
CL_CC = feedback(PI_CC * Gp, 1);
```

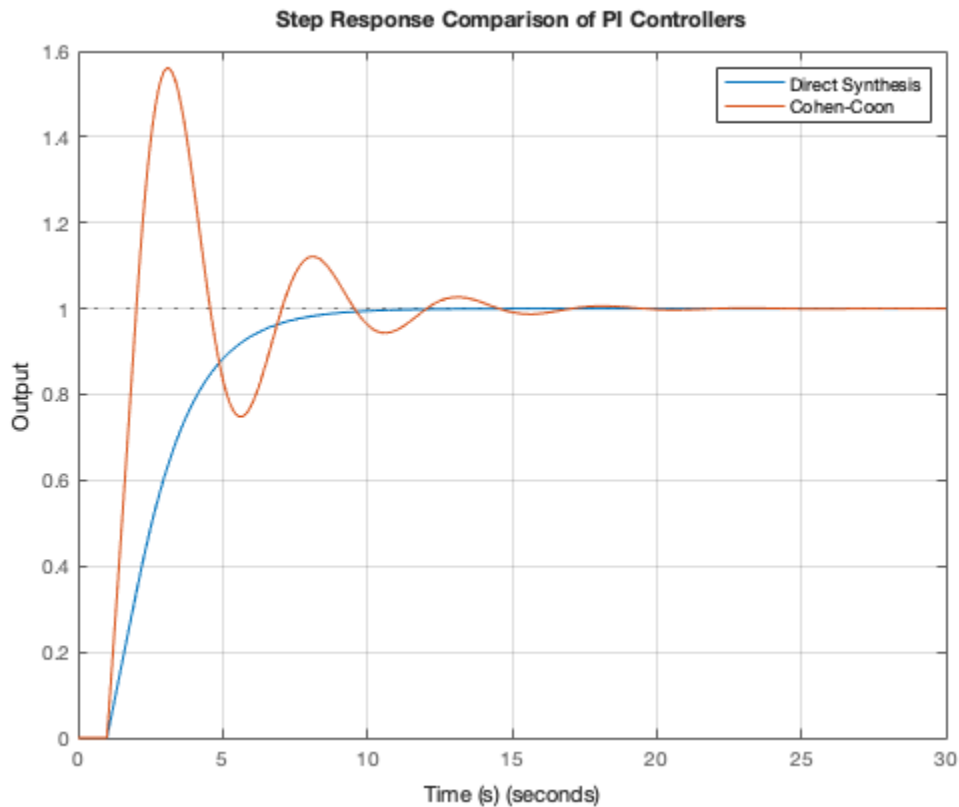
--- Plot Step Responses ---

```
t = 0:0.01:30;
figure;
step(CL_DS, t); hold on;
```

```

step(CL_CC, t);
legend('Direct Synthesis', 'Cohen-Coon');
title('Step Response Comparison of PI Controllers');
xlabel('Time (s)');
ylabel('Output');
grid on;

```



--- Display Controller Parameters ---

```

fprintf('--- Controller Parameters ---\n');
fprintf('Direct Synthesis:      Kc = %.4f, TauI = %.4f\n', Kc_DS, tauI_DS);
fprintf('Cohen-Coon:                Kc = %.4f, TauI = %.4f\n', Kc_CC, tauI_CC);

```

--- Controller Parameters ---

Unrecognized function or variable 'tauI_DS'.

Error in CHE435HW6Q4 (line 39)

```

fprintf('Direct Synthesis:      Kc = %.4f, TauI = %.4f\n', Kc_DS, tauI_DS);
                        ^^^^^^^^

```

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